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Ex.No. 13: V2X Latency Comparison

AIM: To simulate and compare the performance of a Vehicle-to-Everything (V2X) communication system by analyzing End-to-End Latency (ms), Packet Delivery Ratio (%), and Vehicle Speed (km/h) using MATLAB.

Objective:

1. To evaluate the End-to-End Latency (ms) of a V2X communication system under different vehicle operating conditions.
2. To analyze the Packet Delivery Ratio (%) and determine the reliability of V2X communication during data transmission.
3. To study the impact of Vehicle Speed (km/h) on the communication performance and stability of the V2X network.

Procedure:

```
clc;
clear;
close all;
% Vehicle Speed (km/h)
speed = 20:20:120;
% Simulated End-to-End Latency (ms)
latency = [18 16 14 12 10 9];
disp('Vehicle Speed (km/h)  End-to-End Latency (ms)')
disp([speed' latency'])
figure
% ----- Graph 1 -----
subplot(2,1,1)
plot(speed,latency,'-o','LineWidth',2)
```

```
title('Vehicle Speed vs End-to-End Latency')
```

```
xlabel('Vehicle Speed (km/h)')
```

```
ylabel('Latency (ms)')
```

```
grid on
```

```
% ----- Graph 2 -----
```

```
subplot(2,1,2)
```

```
plot(1:length(latency),latency,'-s','LineWidth',2)
```

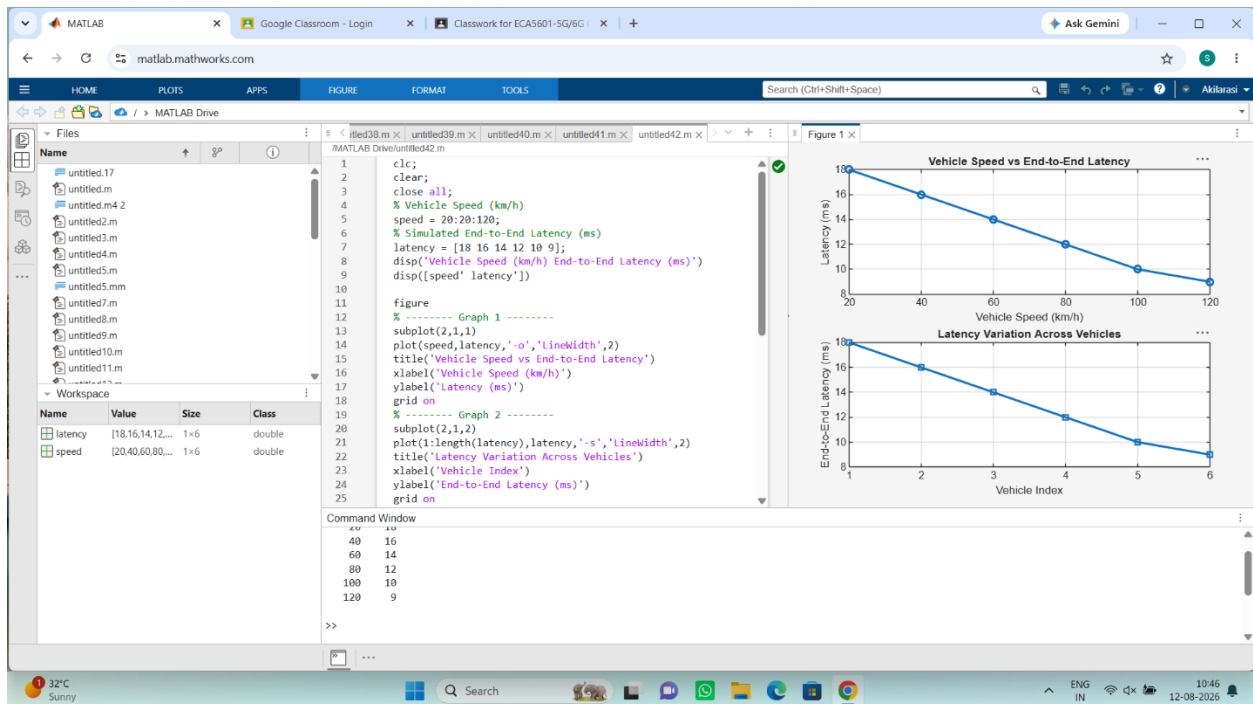
```
title('Latency Variation Across Vehicles')
```

```
xlabel('Vehicle Index')
```

```
ylabel('End-to-End Latency (ms)')
```

```
grid on
```

Output:



Objective 2:

```
clc;

clear;

close all;

% Vehicle Speed (km/h)

speed = 20:20:120; % Simulated Packet Delivery Ratio (%)

PDR = [99 98 97 95 93 90];

disp('Vehicle Speed (km/h)   Packet Delivery Ratio (%)')

disp([speed' PDR'])

figure

% ----- Graph 1 -----

subplot(2,1,1)

plot(speed,PDR,'-o','LineWidth',2)

title('Vehicle Speed vs Packet Delivery Ratio')

xlabel('Vehicle Speed (km/h)')

ylabel('Packet Delivery Ratio (%)')

grid on

% ----- Graph 2 -----

subplot(2,1,2)

plot(1:length(PDR),PDR,'-s','LineWidth',2)

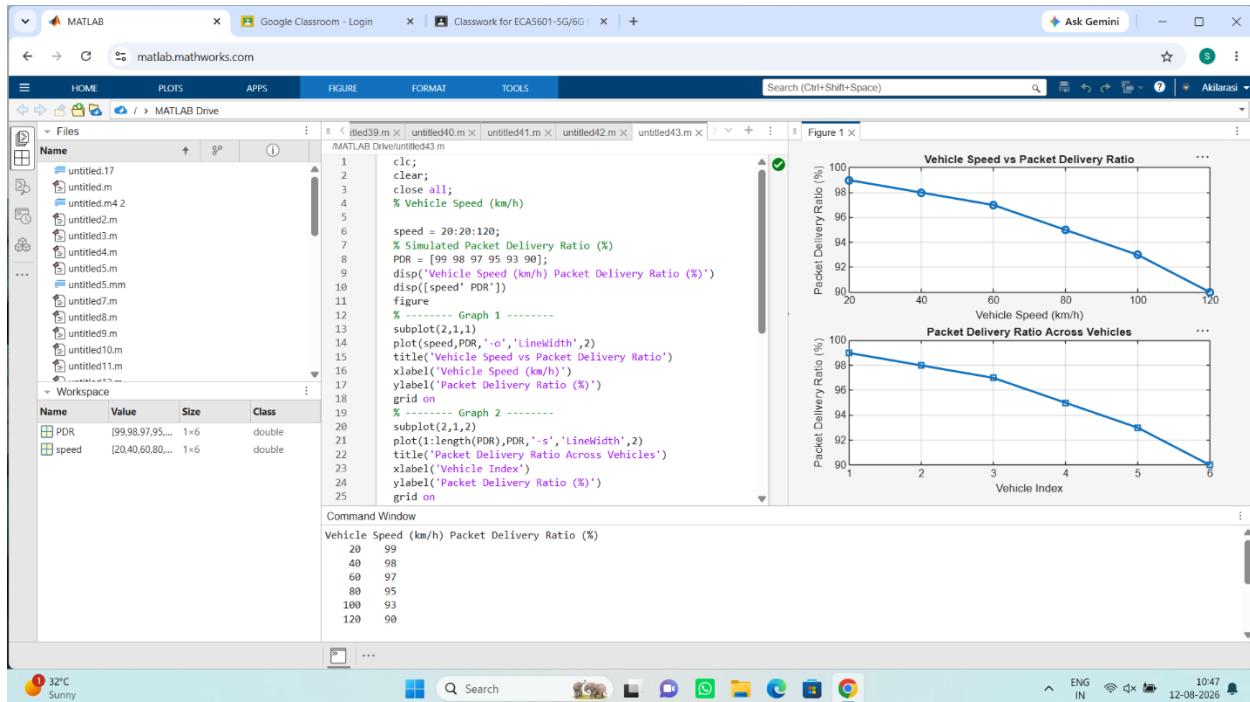
title('Packet Delivery Ratio Across Vehicles')

xlabel('Vehicle Index')

ylabel('Packet Delivery Ratio (%)')

grid on
```

Output:



Objective 3:

`clc; clear; close all;`

`% Vehicle IDs`

`vehicle = 1:6; % Vehicle Speed (km/h)`

`speed = [30 45 60 75 90 105]; % Communication Range (m)`

`range = [280 260 240 220 200 180];`

`disp(['Vehicle Speed(km/h) Communication Range(m)'])`

`disp([vehicle' speed' range'])`

`figure`

`% ----- Graph 1 -----`

`subplot(2,1,1)`

`plot(vehicle,speed,'-o','LineWidth',2)`

`title('Vehicle Speed of Different Vehicles')`

`xlabel('Vehicle Index')`

```
ylabel('Vehicle Speed (km/h)')
```

```
grid on
```

```
% ----- Graph 2 -----
```

```
subplot(2,1,2)
```

```
plot(speed,range,'-s','LineWidth',2)
```

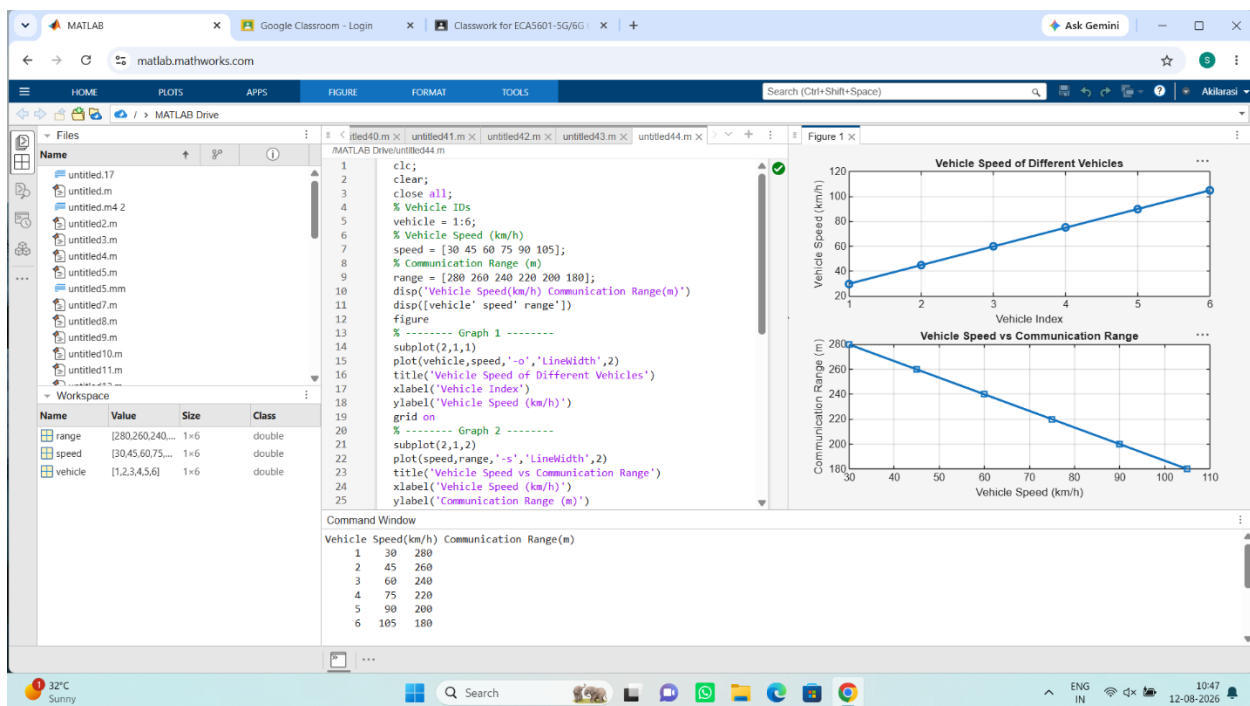
```
title('Vehicle Speed vs Communication Range')
```

```
xlabel('Vehicle Speed (km/h)')
```

```
ylabel('Communication Range (m)')
```

```
grid on
```

Output:



Result:

1. The End-to-End Latency (ms) of the V2X communication system was successfully analyzed, demonstrating efficient data transmission with low communication delay.
2. The Packet Delivery Ratio (PDR %) was evaluated and showed high packet delivery reliability under different vehicle operating conditions.
3. The impact of Vehicle Speed (km/h) on V2X communication was successfully studied, indicating that higher vehicle speeds can influence communication range, latency, and overall network performance.